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PART 1 — BACTERIAL DOUBLING TIME BY POLLUTANT

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PART 2 — ENZYME ACTIVITY (Two-Way ANOVA)

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PART 3 — MOUSE ACTIVITY (Two-Way ANOVA)

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```

STA1004 Assignment 3

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2026-02-21

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PART 1 — BACTERIAL DOUBLING TIME BY POLLUTANT

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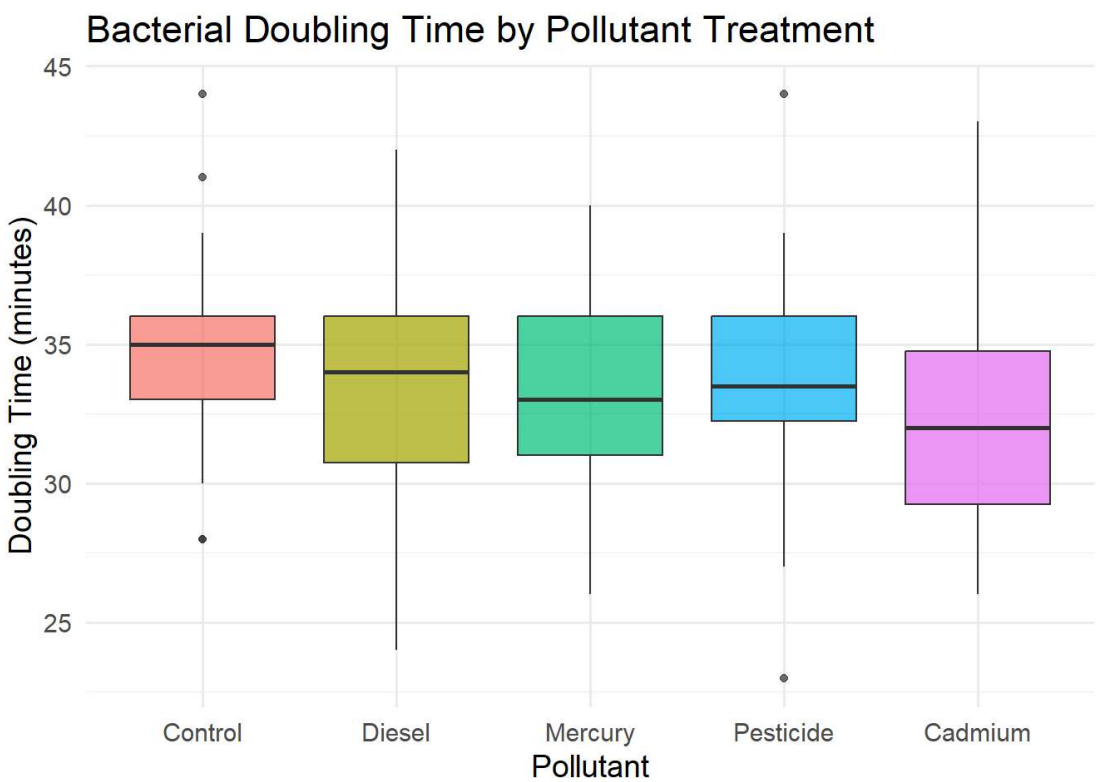
```
bacteria_data <- data.frame(
  pollutant = factor(
    c("Diesel","Control","Diesel","Control","Mercury","Pesticide","Pesticide","Diesel","Mercury","Contro
l","Pesticide","Diesel","Pesticide","Cadmium","Mercury","Diesel","Mercury","Cadmium","Cadmium","Contro
l","Pesticide","Control","Cadmium","Diesel","Pesticide","Diesel","Cadmium","Mercury","Pesticide","Contro
l","Diesel","Mercury","Cadmium","Diesel","Mercury","Pesticide","Mercury","Control","Pesticide","Contro
l","Mercury","Pesticide","Control","Cadmium","Diesel","Mercury","Control","Control","Control","Pesticid
e","Pesticide","Diesel","Mercury","Pesticide","Cadmium","Cadmium","Mercury","Diesel","Diesel","Cadmiu
m","Cadmium","Diesel","Cadmium","Pesticide","Mercury","Pesticide","Cadmium","Cadmium","Cadmium","Cadmiu
m","Cadmium","Cadmium","Control","Diesel","Mercury","Control","Control","Diesel","Mercury","Pesticid
e","Control","Diesel","Pesticide","Mercury","Pesticide","Pesticide","Diesel","Control","Control","Diese
l","Control","Pesticide","Diesel","Control","Cadmium","Diesel","Mercury","Pesticide","Cadmium","Cadmiu
m","Cadmium","Mercury","Diesel","Pesticide","Mercury","Pesticide","Cadmium","Mercury","Mercury","Cadmiu
m","Control","Mercury","Diesel","Control","Pesticide","Cadmium","Pesticide","Control","Diesel","Diese
l","Cadmium","Control","Mercury","Cadmium","Pesticide","Diesel","Diesel","Mercury","Pesticide","Cadmiu
m","Control","Diesel","Control","Control","Mercury","Diesel","Mercury","Cadmium","Control","Pesticid
e","Pesticide","Mercury","Mercury","Control","Pesticide","Mercury","Diesel","Cadmium","Mercury","Contro
l"),
    levels = c("Control","Diesel","Mercury","Pesticide","Cadmium")
  ),
  doubling_time = c(37,28,33,38,39,35,36,36,37,33,32,33,35,29,36,40,31,34,26,35,33,36,30,35,38,35,34,3
9,44,41,34,31,35,24,28,34,36,44,33,35,32,35,35,34,26,32,34,36,37,37,33,29,34,37,32,33,31,35,42,29,37,2
6,35,23,26,34,27,30,32,32,29,43,28,34,35,32,35,25,33,34,34,36,37,29,27,30,35,34,33,42,36,30,37,32,36,3
0,33,33,29,31,28,26,33,33,37,32,31,34,37,30,35,28,35,35,33,39,32,32,38,29,28,30,33,35,31,34,33,40,36,3
5,36,34,36,32,36,30,31,33,35,37,33,32,31,39,39,34,37,31,33,36)
)
```

Boxplot

```
ggplot(bacteria_data, aes(x = pollutant, y = doubling_time, fill = pollutant)) +
  geom_boxplot(alpha = 0.7) +
  theme_minimal(base_size = 14) +
  labs(title = "Bacterial Doubling Time by Pollutant Treatment",
       x = "Pollutant",
       y = "Doubling Time (minutes)") +
  theme(legend.position = "none")
```



Summary Statistics

```
bacteria_data %>%
  group_by(pollutant) %>%
  summarise(
    Mean = round(mean(doubling_time),2),
    SD   = round(sd(doubling_time),2),
    n    = n()
  )
```

```
## # A tibble: 5 × 4
##   pollutant  Mean    SD     n
##   <fct>     <dbl> <dbl> <int>
## 1 Control    34.7  3.32    30
## 2 Diesel     33.6  4.58    30
## 3 Mercury    33.1  3.66    30
## 4 Pesticide  33.9  3.79    30
## 5 Cadmium    32.2  3.73    30
```

One-Way ANOVA

```
bartlett.test(doubling_time ~ pollutant, data = bacteria_data)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  doubling_time by pollutant
## Bartlett's K-squared = 3.3003, df = 4, p-value = 0.5089
```

```
model1 <- aov(doubling_time ~ pollutant, data = bacteria_data)
summary(model1)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## pollutant     4    102   25.49   1.728  0.147
## Residuals   145   2140   14.76
```

```
TukeyHSD(model1)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = doubling_time ~ pollutant, data = bacteria_data)
##
## $pollutant
##                         diff       lwr       upr     p adj
## Diesel-Control    -1.1666667 -3.906457 1.5731240 0.7648918
## Mercury-Control   -1.6000000 -4.339791 1.1397907 0.4914231
## Pesticide-Control -0.8666667 -3.606457 1.8731240 0.9060743
## Cadmium-Control   -2.5000000 -5.239791 0.2397907 0.0916745
## Mercury-Diesel    -0.4333333 -3.173124 2.3064574 0.9923570
## Pesticide-Diesel   0.3000000 -2.439791 3.0397907 0.9981596
## Cadmium-Diesel    -1.3333333 -4.073124 1.4064574 0.6640119
## Pesticide-Mercury  0.7333333 -2.006457 3.4731240 0.9468717
## Cadmium-Mercury   -0.9000000 -3.639791 1.8397907 0.8936626
## Cadmium-Pesticide -1.6333333 -4.373124 1.1064574 0.4702022
```

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PART 2 — ENZYME ACTIVITY (Two-Way ANOVA)

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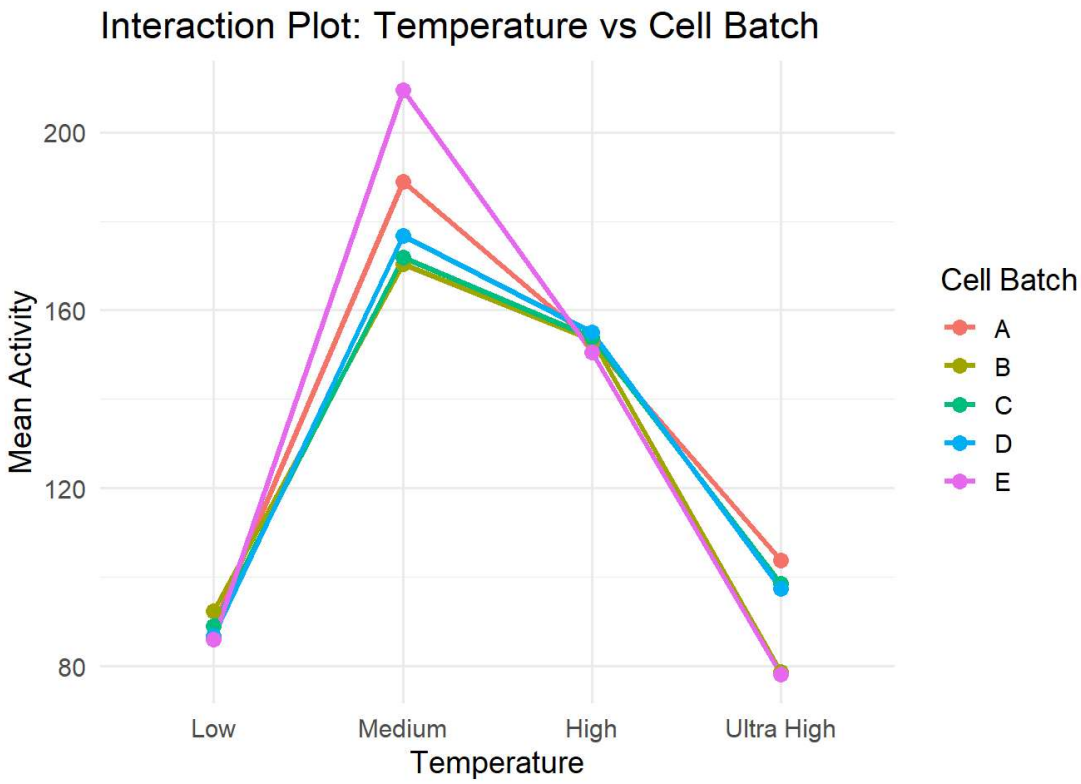
```
enzyme_data <- data.frame(
  temp = factor(rep(c("Low","Medium","High","Ultra High"), each = 50),
                levels = c("Low","Medium","High","Ultra High")),
  cellbatch = factor(rep(rep(LETTERS[1:5], each = 10), 4)),
  activity = c(
    69,80,88,87,85,100,105,104,83,89,102,89,79,96,96,84,96,100,85,96,
    81,98,75,79,106,96,91,92,85,87,95,75,62,102,81,92,90,82,86,103,
    74,73,95,76,80,92,87,94,100,89,
    186,192,233,213,191,208,162,174,168,161,
    156,185,142,225,197,168,125,184,216,105,
    151,164,116,188,242,166,158,197,168,168,
    228,251,152,202,106,217,78,179,120,234,
    191,217,266,225,209,190,216,197,207,177,
    145,141,144,177,150,168,152,163,146,141,
    164,147,163,130,160,145,140,177,147,161,
    163,148,167,149,126,167,143,159,140,177,
    166,140,148,156,147,156,178,141,162,157,
    153,150,173,146,149,165,127,128,163,152,
    99,34,117,66,57,113,178,193,53,127,
    4,141,64,38,153,71,115,62,131,8,
    6,204,57,115,79,108,28,143,167,79,
    54,148,88,87,186,71,132,98,-22,133,
    24,20,79,117,86,217,47,55,27,110
  )
)
```

Interaction Plot

```
ggplot(enzyme_data, aes(x = temp, y = activity,
                        color = cellbatch, group = cellbatch)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  theme_minimal(base_size = 14) +
  labs(title = "Interaction Plot: Temperature vs Cell Batch",
       x = "Temperature",
       y = "Mean Activity",
       color = "Cell Batch")
```



Two-Way ANOVA

```
model2 <- aov(activity ~ temp * cellbatch, data = enzyme_data)
summary(model2)
```

```
##                 Df Sum Sq Mean Sq F value Pr(>F)
## temp             3 329651  109884  87.058 <2e-16 ***
## cellbatch        4   2149     537   0.426  0.790
## temp:cellbatch  12  14575    1215   0.962  0.487
## Residuals      180 227195    1262
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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PART 3 — MOUSE ACTIVITY (Two-Way ANOVA)

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Create Data

```
mouse_data <- data.frame(
  time = factor(rep(c("morning","midday","afternoon"), each = 21),
                levels = c("morning","midday","afternoon")),

  light = factor(rep(rep(c("low","medium","bright"), each = 7), times = 3),
                 levels = c("low","medium","bright")),

  activity = c(
    14.1,5,14.7,7.3,8.7,5.3,13.3,
    14.5,10.8,11.4,11.7,8.6,10.1,16.8,
    10.1,9.6,11.7,19.2,21.8,13,2.7,

    16,18.7,9.1,8.9,11.9,10.6,18.8,
    6.6,7.8,14.3,10.3,12,21.9,17.9,
    13.9,18.6,8.3,21.1,28.2,11.5,14.5,

    12.5,16.1,9.3,9.3,12,12.4,9.4,
    9.1,20.8,16.7,13.7,17.6,11.9,4.8,
    20.6,19.4,12.4,14.9,14.2,9.5,19.7
  )
)

dim(mouse_data)
```

```
## [1] 63  3
```

```
table(mouse_data$time, mouse_data$light)
```

```
##
##             low medium bright
##   morning     7      7      7
##   midday      7      7      7
##   afternoon   7      7      7
```

```
model3 <- aov(activity ~ time * light, data = mouse_data)
summary(model3)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## time         2   94.9   47.46   2.010 0.1438
## light        2  124.8   62.40   2.643 0.0803 .
## time:light   4   23.3    5.84   0.247 0.9101
## Residuals   54 1274.8   23.61
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
bartlett.test(activity ~ interaction(time, light), data = mouse_data)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  activity by interaction(time, light)
## Bartlett's K-squared = 9.1872, df = 8, p-value = 0.3267
```

```
shapiro.test(residuals(model3))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(model3)
## W = 0.98263, p-value = 0.5157
```

```
TukeyHSD(model3)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = activity ~ time * light, data = mouse_data)
##
## $time
##                          diff        lwr      upr     p adj
## midday-morning     2.8809524 -0.7326773 6.494582 0.1424686
## afternoon-morning  2.1857143 -1.4279154 5.799344 0.3192080
## afternoon-midday  -0.6952381 -4.3088678 2.918392 0.8885069
##
## $light
##                   diff        lwr      upr     p adj
## medium-low    1.233333 -2.3802963 4.846963 0.6907917
## bright-low    3.404762 -0.2088678 7.018392 0.0686931
## bright-medium 2.171429 -1.4422011 5.785058 0.3238745
##
## $`time:light`
##                                          diff        lwr       upr     p adj
## midday:low-morning:low             3.65714286  -4.733462 12.047748 0.8898021
## afternoon:low-morning:low          1.80000000  -6.590605 10.190605 0.9986905
## morning:medium-morning:low         2.21428571  -6.176319 10.604890 0.9944692
## midday:medium-morning:low          3.20000000  -5.190605 11.590605 0.9455499
## afternoon:medium-morning:low       3.74285714  -4.647748 12.133462 0.8765154
## morning:bright-morning:low         2.81428571  -5.576319 11.204890 0.9742193
## midday:bright-morning:low          6.81428571  -1.576319 15.204890 0.2010923
## afternoon:bright-morning:low       6.04285714  -2.347748 14.433462 0.3450089
## afternoon:low-midday:low          -1.85714286 -10.247748  6.533462 0.9983642
## morning:medium-midday:low         -1.44285714  -9.833462  6.947748 0.9997396
## midday:medium-midday:low          -0.45714286  -8.847748  7.933462 1.0000000
## afternoon:medium-midday:low        0.08571429  -8.304890  8.476319 1.0000000
## morning:bright-midday:low         -0.84285714  -9.233462  7.547748 0.9999957
## midday:bright-midday:low           3.15714286  -5.233462 11.547748 0.9495092
## afternoon:bright-midday:low        2.38571429  -6.004890 10.776319 0.9909193
## morning:medium-afternoon:low       0.41428571  -7.976319  8.804890 1.0000000
## midday:medium-afternoon:low        1.40000000  -6.990605  9.790605 0.9997921
## afternoon:medium-afternoon:low     1.94285714  -6.447748 10.333462 0.9977512
## morning:bright-afternoon:low       1.01428571  -7.376319  9.404890 0.9999821
## midday:bright-afternoon:low        5.01428571  -3.376319 13.404890 0.5962744
## afternoon:bright-afternoon:low     4.24285714  -4.147748 12.633462 0.7821353
## midday:medium-morning:medium       0.98571429  -7.404890  9.376319 0.9999856
## afternoon:medium-morning:medium    1.52857143  -6.862033  9.919176 0.9996007
## morning:bright-morning:medium      0.60000000  -7.790605  8.990605 0.9999997
## midday:bright-morning:medium       4.60000000  -3.790605 12.990605 0.7001559
## afternoon:bright-morning:medium    3.82857143  -4.562033 12.219176 0.8623409
## afternoon:medium-midday:medium     0.54285714  -7.847748  8.933462 0.9999999
## morning:bright-midday:medium      -0.38571429  -8.776319  8.004890 1.0000000
## midday:bright-midday:medium        3.61428571  -4.776319 12.004890 0.8961087
## afternoon:bright-midday:medium     2.84285714  -5.547748 11.233462 0.9725979
## morning:bright-afternoon:medium   -0.92857143  -9.319176  7.462033 0.9999910
## midday:bright-afternoon:medium     3.07142857  -5.319176 11.462033 0.9568187
## afternoon:bright-afternoon:medium  2.30000000  -6.090605 10.690605 0.9928680
## midday:bright-morning:bright       4.00000000  -4.390605 12.390605 0.8314103
## afternoon:bright-morning:bright    3.22857143  -5.162033 11.619176 0.9427947
## afternoon:bright-midday:bright    -0.77142857  -9.162033  7.619176 0.9999979
```

```r
model_time_only    <- aov(activity ~ time, data = mouse_data)
model_light_only   <- aov(activity ~ light, data = mouse_data)
model_additive     <- aov(activity ~ time + light, data = mouse_data)
model_interaction  <- aov(activity ~ time * light, data = mouse_data)

AIC(model_time_only,
    model_light_only,
    model_additive,
    model_interaction)
```

```
##                   df      AIC
## model_time_only    4 383.1791
## model_light_only   4 381.8423
## model_additive     6 381.3962
## model_interaction 10 388.2531
```